

NBA Shot Outcomes by Player Tracking Data

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Model and Reasoning

A Logistic Regression model was used to properly train on the training data, and apply findings to the testing data. A Logistic regression model was chosen due to the outcome of the shot being binary (make or miss), and having a probability calculated between 0 and 1.

Features

When choosing between the features in the dataset making sure no overlapping categories and unnecessary data was included. I sorted the variables into 4 categories shown in the table below, and selected 5 variables I decided were most valuable, and did not overlap. I then tested them using the regression model, and all features were 3 stars, showing their value in influencing the outcome.

contestState	gameSituation	shotScenario	unnecessary
numContesters**	gameState**	shotType**	month
distcont (1-4)	shotClock***	distance**	
contested	dribblesBefore	location (x&y)*	
closestDefDist*		three	
closestDefApproach		shooterSpeed	

Legend

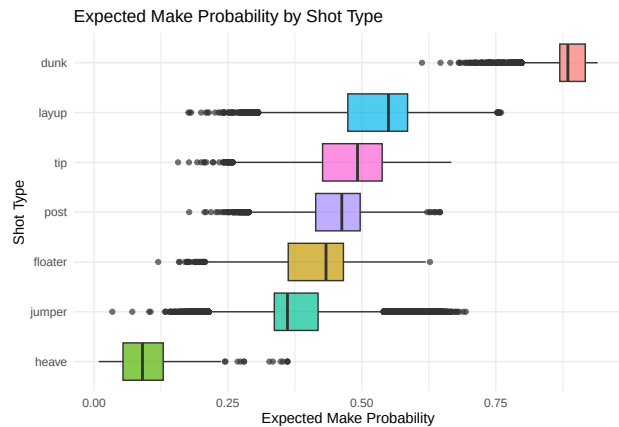
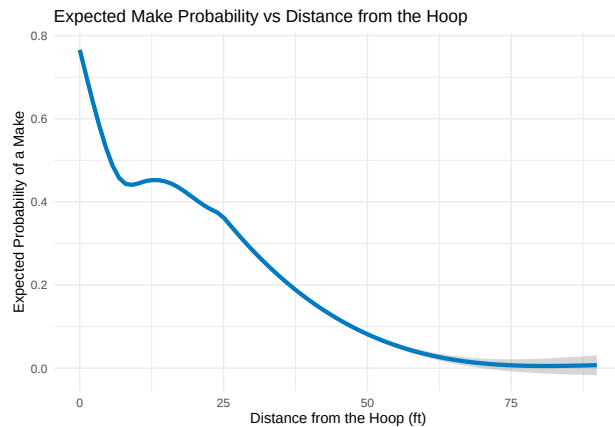
* - Maybe

** - Yes

Number of contesters is the best feature for contestState, as it captures the boolean value of contested, but keeps data which is relevant, and by my tests, closestDefDist was less impactful.

gameSituation can be understood by shotClock and gameState, giving the proper scenario of the game, and time remaining, allowing to understand the nature of the game, beyond distance of the shot and contesting.

shotScenario can be best captured by shotType and distance, showing what kind of shot is being taken, as a jumper vs a post shot from the same distance have much different projected outcomes as shown in the graphs below



Takeaways

With the data given, the model is able to create a value of probability for a shot made, and given more time, there are definitely changes I would make. The data on locationX and locationY reveal a lot, but I was unable to utilize them appropriately. Using them to show areas of the court which indicate an easier shot could be useful. For example, a mid range jumper from the corner is inherently a lower percentage shot compared to one from the same distance at the elbow, due to the backboard being available.

I also believe other data points could be used for finding a more optimal probability. Many teams go on runs to accumulate points, and momentum is a heavy factor in gaining and retaining leads. A stat such as uninterrupted points, or time since timeout could add more value to the dataset to better identify the state of the game.

Reflection

This project is one I had a lot of fun with. Working with data that I have passion for makes it feel less like a task, and more like a hobby. Using a logistic Regression model was new to me for this project, so learning and being adaptive to a new style was something I enjoyed, and take pride in being a strong suit of mine.